

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10 – REAL NUMBERS (1.1) – FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Think back to Sister Wanjiku at the Mathare North Health Centre in Nairobi. Her calculator gave a neat answer for the average temperature, $(36.8 + 37.0 + 37.5) \div 3$, but a never-ending decimal for the diagonal of the 1 m square waiting-room tile. In the space under each prompt, write your own explanation using evidence from our lessons. Use correct vocabulary (rational, irrational, p/q , terminating, repeating, reciprocal). Show your working for any calculation. Give reasons for your claims, not just answers. Write in full sentences.

Part 1: What kind of numbers did Sister Wanjiku record?

Sister Wanjiku recorded body temperatures of 36.8 °C, 37.0 °C and 37.5 °C. She also recorded blood pressure readings of 120/80 and 135/90, and shoe sizes of 40, 41 and 43. (a) State what it means for a number to be rational. (b) Write each of the three temperatures as a fraction p/q with p and q integers and $q \neq 0$. (c) Explain whether the shoe sizes also belong to the set of rational numbers, and say which other number sets they belong to.

Write your answer here:

Part 2: Why did the average temperature 'end neatly'?

Sister Wanjiku calculated $(36.8 + 37.0 + 37.5) \div 3$. (a) Work out the answer, showing each step. (b) Write your answer as a fraction p/q . (c) Using what you learned about operations on rational numbers, explain what you could say about the type of number the answer would be before using the calculator, and give your reasons.

Write your answer here:

Part 3: Why does the tile diagonal go on forever?

The waiting-room floor tile is a square with side 1 m. (a) Use the Pythagorean theorem to write an expression for the diagonal length. (b) Explain what is different about this number compared with Sister Wanjiku's average temperature. (c) Sister Wanjiku's calculator shows 1.41421356... A classmate says, 'The calculator display stops, so the decimal must terminate.' Explain why this reasoning is not safe. (d) Name one other irrational number we met in this topic, and explain how $22/7$ differs from π .

Write your answer here:

Part 4: Reciprocals

(a) Define the reciprocal of a non-zero real number x . (b) Find the reciprocal of 37.1 as a fraction. (c) Explain why the reciprocal of $\sqrt{2}$ is irrational. (d) Which number has no reciprocal, and why?

Write your answer here:

Part 5: Answering the driving question

Driving question: Why do some numbers from our everyday Kenyan lives 'end neatly' while others go on forever, and how does knowing the difference help us make better decisions? Write a clear answer of 5–7 sentences that uses Sister Wanjiku's data (the average temperature and the tile diagonal). Include one example from health, construction or commerce showing why knowing whether a number is rational or irrational matters.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Classification of numbers (rational vs irrational)	Defines rational and irrational numbers accurately, writes every health-centre number as p/q correctly, and places numbers in $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$ with correct reasons.	Defines the terms mostly correctly and writes most numbers as p/q , with minor errors in set placement or reasons.	Definitions are incomplete or incorrect, or learner cannot write numbers as p/q or place them in number sets.
Use of closure and operations on rational numbers	Calculates 37.1 correctly with clear steps and uses closure under addition and division to justify why the result is rational.	Calculates the average correctly and mentions closure, but the justification is incomplete.	Calculation contains errors or the learner gives no reason for why the result ends neatly.
Explaining irrational numbers and calculator use	Explains why $\sqrt{2}$ is irrational, why a truncated display does not prove a decimal terminates, and distinguishes 22/7 from π accurately.	Identifies $\sqrt{2}$ as irrational and notes the calculator limitation, but the reasoning or the 22/7 versus π distinction is partly unclear.	Treats the calculator display as exact or cannot explain why $\sqrt{2}$ differs from the average temperature.
Reciprocals and mathematical reasoning	Defines reciprocals, finds 10/371 correctly, gives a correct proof by contradiction for $1/\sqrt{2}$, and states that zero has no reciprocal.	Finds correct reciprocals and states $1/\sqrt{2}$ is irrational with a partial argument.	Reciprocal definition or calculations are incorrect and no reasoning is offered.
Final explanation linked to Kenyan context and decisions	Gives a coherent answer to the driving question using Sister Wanjiku's data and a realistic decision example.	Answers the driving question using the data, with a weak or general decision example.	Answer is vague, not linked to the data, or gives no decision example.

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